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Adobe: GenAI Opportunity or Threat?

It was early-December 2022 and Shantanu Narayen, the CEO of Adobe, awaited his team in the boardroom atop the company's San Jose, California headquarters. Glancing at his phone he caught sight of headlines about the astonishing speed of adoption of OpenAI's recently released ChatGPT. The tool's unprecedented ability to generate human-like text from simple prompts had rattled the tech world and beyond. The breakthrough came on the heels of a wave of new generative AI (GenAI) image model releases over the summer. OpenAI's DALL-E 2, and the models of smaller players like Midjourney and Stability AI had captivated the public with their magic-like ability to render visual pictures from text prompts. Narayen was keenly aware of these developments and the potential disruption they could bring to Adobe's core business.

Over its 40-year history, Adobe had built software products to empower individuals to create, collaborate, and bring their visions to life. It was a global leader in digital media and marketing solutions, anchored by its Creative Cloud (including Photoshop, Illustrator, and Premiere Pro), Document Cloud (Acrobat and PDF services), and Experience Cloud (customer experience management and analytics). It served a range of customers from the world's largest enterprises, to small and medium sized business, to creative professionals in marketing, advertising, and media, to influencers, hobbyists, and students. Roughly one decade earlier, the company had successfully shifted to a subscription model, bolstering recurring revenue and deepening customer engagement.

While Adobe's performance had climbed steadily in recent years—its market capitalization surpassed \$315 billion in December 2021—the stock was down by roughly 50% year over year. Analysts pointed to macroeconomic conditions, a post-pandemic slowdown in enterprise software purchasing, and increased competition. (See **Exhibit 1**.)

While the world was buzzing with excitement about developments in GenAI models, the reaction of creative professionals—Adobe's primary customer base—was in large part that of distrust, angst, and a concern for their livelihoods. For Narayen the tension was clear: Did this technology represent a threat or an opportunity for Adobe's core? He and his team would need to make some tough decisions with respect to this burgeoning technology.

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Overview of Adobe's History

In 1982, Dr. John Warnock and Dr. Charles Geschke left Xerox PARC to found Adobe, named after a creek near Warnock's home.¹ They launched PostScript, a computer language that translated digital text and images from a computer screen onto the printed page, to solve for file and printer incompatibility issues.² Adobe went public in 1986,³ and through the 1980s and early 1990s, it entered the consumer software market by building up and acquiring applications such as Adobe Illustrator, Photoshop, and PageMaker,⁴ which were licensed to creative professionals like graphic and web designers, videographers, and photographers. In 1993, Adobe created the Portable Document Format (PDF), allowing anyone to send an electronic document to anyone else and print it while preserving the page layout.⁵ By the late 1990s, it had grown to become a major player in the software industry.⁶

Over the 2000s, Adobe provided creative professionals with a comprehensive toolkit and promoted cross-application workflows. In 2003, it launched the Creative Suite and bundled popular products such as Photoshop, InDesign, and Acrobat.⁷ In 2005, seeking to expand its web development capabilities, Adobe acquired Macromedia—known for software like Flash and Dreamweaver—thereby broadening its reach in animation and web development. That same year, Adobe introduced Stock Photos, allowing users to search, manage, and buy royalty-free stock images.⁸ At launch, Stock Photos offered access to over 230,000 photos and illustrations.⁹ Adobe later augmented its stock image library in 2014 by acquiring Fotolia, a marketplace of over 35 million stock images and videos.¹⁰

Narayan assumed the role of CEO in late 2007, just before the 2008 financial crisis put pressure on Adobe as global corporate budgets for software contracted. The company's revenue declined and approximately 600 employees—about 8% of its workforce—were laid off to reduce costs.¹¹ Yet it was during Narayan's early tenure that Adobe expanded into enterprise marketing solutions by acquiring Omniture, an online marketing and web analytics company whose tools helped customers track and manage digital marketing efforts. Over time, these offerings grew to become another pillar of company revenues separate from creative and document software.

A Bold Shift to a Cloud-Based Subscription Model

The Omniture acquisition also spurred Adobe to rethink its monetization model. Previously, Adobe's creative software products were sold through computer software resellers and large-box retailers like Staples and Best Buy. By contrast, Omniture sold its products through a cloud-based subscription model, which had the advantage of smoother revenue streams compared to Adobe's sudden spikes when products were launched.¹² Narayan was eager to transition to a subscription model. He expected that the new cloud-based subscription might bring in new users given the lower up-front costs while also increasing the satisfaction of existing customers who would receive updates continuously online. Thus, despite pushback from longstanding users, Adobe migrated its software from a packaged sales model to a cloud-based subscription model over the next several years.¹³ The move, which had been considered risky at the outset, paid off.

The Next Chapter: Building AI into Adobe's Products

Adobe had long invested in research on cutting-edge technology, launching its internal innovation lab in 1988. An early example of its AI work was Photoshop's "red-eye" correction feature. After the Omniture acquisition, the company expanded its analytics-based AI efforts, applying predictive modeling to marketing technology and integrating insights with content creation. By 2016, Adobe's broader AI initiatives coalesced under Sensei, a framework that unified machine learning and data

analytics under one umbrella so it could be infused across the product suite.¹⁴ Adobe voiced its stance on AI: “Sensei will amplify human intelligence, not replace it.”¹⁵

The company was keenly aware of the ethical implications of AI tools. At its 2016 Adobe MAX conference, for instance, a live demo of a prototype called “Photoshop for voice” showed how easily recorded speech could be manipulated. While Adobe had already been working on watermarking recordings for authenticity, this event spurred further investments to foster ethical AI use in subsequent years. In 2019, Adobe launched its AI Ethics Board with a comprehensive review program for all its AI products. That same year, Adobe, along with the New York Times Group and Twitter, launched the Content Authenticity Initiative, a global, cross-industry coalition later joined by The Wall Street Journal, Nvidia, Microsoft, Nikon, Canon, Sony, Universal Music Group, and over 4,000 others to develop tools for digital content attribution that allowed creators and consumers to trace the origins of content—helping combat deepfakes and misinformation. To further advance a shared standard, Adobe became a founding member of Coalition for Content Provenance and Authenticity (C2PA), an open, global standards organization that developed the technical specifications behind a digital nutrition label for content, Content Credentials. In 2021, Adobe introduced Content Credentials into its product suite, letting creators attach attribution to their work.¹⁶ “The focus was on provenance, not detection forensics,” noted one executive. C2PA came to include the BBC, Intel, and Sony, among others.¹⁷

Adobe’s Business in Late 2022

Adobe was a global business with just over half of its roughly 30,000 employees based in the United States. It was long recognized as a company with a positive culture, being named among *Fortune’s* Best Workplaces in Tech.¹⁸ Many senior executives had decades-long tenures, and in several cases had “boomeranged” back to Adobe after departing for jobs elsewhere. Approximately 60% of its revenue came from the Americas, one-quarter from EMEA, and the balance from the Asia Pacific region. Adobe’s Q4 2022 ended with record-breaking revenue of \$4.53 billion, representing 10% year over year growth. The company defined its business across two strategic areas of investment – Digital Media and Digital Experience. (See **Exhibit 2** for segment breakdown and financial statements.)

Digital Media: Unleashing Creativity and Accelerating Document Productivity

The Digital Media segment, overseen by David Wadhwani, brought in nearly three-quarters of Adobe’s revenue. It offered creative professionals and consumers a platform for expression across imaging, photography, video, design, illustration, and 3-D—areas of substantial growth given the increasing importance of digital content to the global economy.

At its heart was the Creative Cloud business, which brought in \$2.68 billion in revenue in Q4, and which enabled a broad spectrum of users to create and collaborate across desktop, tablet, and mobile devices. Its primary products were available in bundles and as point solutions, serving everyone from seasoned professionals to hobbyists. Professional creators often integrated several apps into a seamless workflow, for example shooting raw footage and editing in Lightroom, refining in Photoshop, and then doing layout in InDesign. While hobbyists gravitated toward simpler template-driven tools such as Adobe Express for quick projects like invitations, social media posts, or school assignments. Along with Express, Adobe’s video segment (Premiere Pro, After Effects, and the Frame.io collaboration platform) and 3-D content tool (Substance) represented notable growth opportunities. (See **Exhibit 3** for list of products and services and customer breakdown.)

Meanwhile, the Document Cloud business generated \$629 million for Q4, growing more than 20% year on year. Document Cloud represented Adobe's initiatives to help customers migrate away from paper-based workflows and toward digitized document experiences. Adobe primarily employed a product-led growth strategy in this area. Its products included apps to help create, convert, edit, and share PDFs across workflows and devices. Their aim was to help customers increase their productivity through collaboration enabled by the ability to share, edit, track, and e-sign digital documents. The Document Cloud was built around the Adobe Acrobat family of products. To date, there were over 2.6 billion cumulative downloads of Acrobat and Reader around the world. Particular areas of growth were Acrobat Web and Acrobat Sign. (See **Exhibit 4** for list of products and services and customer breakdown.)

Digital Experience: Powering Digital Business

Adobe's Digital Experience segment, launched in 2012 and overseen by Anil Chakravarthy, represented the company's second pillar of growth. As institutions worldwide undertook digital transformations, they required new platforms and tools to engage with their customers from analytics to commerce. This field, known as customer experience management, used digital marketing tools along with customer data platforms (CDP) that collected real-time data across multiple systems.

Adobe Experience Cloud translated real-time customer data insights into profile-based, personalized digital experiences across multiple channels at scale. The underlying Adobe Experience Platform leveraged AI and offered the open architecture and ecosystem to support transformation of user data from Adobe solutions and third-party software into actionable customer profiles. Experience Cloud held the top position in Gartner's Magic Quadrant for Digital Experience for over eight years, since the inception of the category. Customers of Digital Experience included advertisers, marketers, merchants, and developers. Many of these users also used Adobe's Digital Media products, which made it easier for them to make, manage, measure, and monetize their content with coordinated workflows.¹⁹ Digital Experience solutions were sold through a direct sales force as well as through partners such as system integrators and software vendors. Customers were large enterprises such as Delta Airlines, The Coca-Cola Company, Disney, Morgan Stanley, and NASA.²⁰ Experience Cloud was the hub of a suite of applications and services and it brought in \$1.4 billion revenue in Q4. (See **Exhibit 5** for list of products and customer breakdown.)

Beyond individual products, Adobe positioned itself as an enabler of end-to-end workflows. Its tools were designed to connect the entire content supply chain. This integration allowed creative and marketing professionals at enterprises of all sizes to move seamlessly across Creative Cloud and Experience Cloud applications to create and unify brand assets, campaigns, and analytics in one cohesive workflow. (See **Exhibit 6** for sample content supply chain workflow.)

Competitive Landscape

While Adobe's Creative Cloud suite had long been the industry standard with creative professionals, the company faced competition primarily from the lower-end of the market. One of the most well-known competitors was ten-year-old, Australia-based Canva. The company had raised over \$540 million in private investments and as of late 2022, its valuation was reportedly \$26 billion.²¹ Canva offered an accessible, user-friendly online design platform that featured a vast library of templates, design elements, and easy-to-use tools. These offerings appealed to the growing set of solopreneurs and influencers. Having reached over 100 million monthly active users across over 190 countries,²² it was a popular choice for individuals and businesses seeking to create visual content, but who lacked extensive design experience. Canva's freemium model and intuitive interface placed it in competition

with Adobe Express. Meanwhile, another fast-growing challenger was San Francisco-based Figma, founded in 2012. By mid-2021, it had raised approximately \$333 million with its most recent round at a reported valuation of around \$10 billion.²³ Figma offered a collaborative, cloud-based design platform tailored for UI/UX professionals, featuring real-time co-editing and intuitive prototyping tools. On September 15, 2022, Adobe announced a plan to acquire Figma. (See **Exhibit 7a** for other competitors.)

On the Digital Experience side, Adobe faced competition from several tech incumbents offering marketing automation, analytics, and customer engagement tools. Rivals included both point solutions, like Braze, and other end-to-end marketing platforms. One of the most prominent was Salesforce, founded in 1999 and headquartered in San Francisco. Originally focused on customer relationship management (CRM) software, Salesforce had significantly broadened its scope—by September 2022, it included cloud-based marketing, data analytics, and collaboration tools. Meanwhile, Google Analytics provided a widely adopted platform for website traffic measurement, user behavior analysis, and data-driven marketing insights. Its integration with Google’s advertising ecosystem meant that many businesses looking for a seamless analytics and ad-buying pipeline saw it as an alternative to Adobe’s analytics solutions. (See **Exhibit 7b** for other competitors.)

Select Developments in GenAI Image Generation

Early experimentation with computer-generated visuals dated back to the 1960s, when cognitive neuroscientists and computer engineers at Bell Labs toyed with ASCII art to show that machines could do more than perform calculations. In the 1970s, British painter Harold Cohen developed his “AARON” system at Stanford’s Artificial Intelligence Lab, using algorithms to produce original drawings and paintings. (See **Exhibit 8** for examples.) Over subsequent decades, researchers explored neural networks—loosely modeled on the human brain—within academic or private labs.²⁴ A key development came in 1997, when teams demonstrated that neural networks could read handwritten digits in postal zip codes, paving the way for later creative applications.²⁵

By 2010, the ImageNet Large Scale Visual Recognition Challenge at Stanford accelerated computer-vision research by offering a massive, labeled dataset. Two years later, Geoffrey Hinton’s team at the University of Toronto reduced classification errors with deep convolutional neural networks, an innovation that earned Hinton a Turing Award. His foundational work on neural network models included the Boltzmann machine, which could both classify images and generate new examples based on learned patterns. In 2014, Ian Goodfellow—then a doctoral student at the University of Montreal—introduced Generative Adversarial Networks (GANs), enabling AI to generate images so realistic they could fool human observers. Around the same time, Google’s DeepDream drew public fascination with its hallucinatory imagery achieved through pattern amplification, proving that neural networks could reimagine existing visuals.²⁶ Meanwhile, researchers at Stanford were pioneering diffusion models—adding and removing noise to produce high-fidelity results. Neural Style Transfer, introduced in 2016, further showcased AI’s creative potential by blending styles of famous artists with personal photographs.²⁷

A major shift arrived in 2017 with transformer architecture, as described in Google’s paper “Attention is All You Need.”²⁸ Transformers quickly revolutionized both visual and language models setting the stage for the series of large language models (LLMs) in the ensuing years. In 2021, San Francisco-based OpenAI, the non-profit that had been infused by \$1 billion from Microsoft two years prior, made waves when it introduced a text-to-image model called DALL-E (named for Salvador Dalí and Pixar’s WALL-E). Initially built on a transformer-based approach with a discrete variational

autoencoder, DALL-E 2 had evolved into a diffusion-based model by April 2022, capable of producing more realistic visuals like the viral “armchair in the shape of an avocado.”²⁹ (See **Exhibit 9** for sample.) Within five months, OpenAI reported that more than 1.5 million users had generated over 2 million images per day with DALL-E 2.³⁰ Google followed with its own text-to-image model, Imagen, in May 2022.³¹

By late 2022 several other companies gained traction. Midjourney—a San Francisco-based research lab of roughly a dozen employees that had not raised private VC funding—had introduced a fourth version of its text-to-image service to rave reviews.³² Its Discord server reached over one million users within six months of its release (See **Exhibit 10** for sample Midjourney images).^{33,34} Stability AI, a London-based startup that had raised \$110 million in private funding, released its open-source Stable Diffusion model, enabling anyone with a capable GPU^a to create high-fidelity AI-generated art. Its model was reportedly downloaded over 200,000 times within months.

On November 30, 2022, OpenAI launched ChatGPT, a conversational interface built on top of the GPT 3.5 LLM. Its user base soared to one million within the first five days—outpacing previous tech adoption curves. While ChatGPT was focused on text generation rather than images, the media spotlight on its uncanny capabilities broadened public interest in GenAI overall.

Reactions to Generative AI Images

Tech enthusiasts and art hobbyists began posting GenAI text-to-image art on social media, often marveling at the seemingly magical capabilities. Some early experimenters embraced the technology, applauding the ease with which they could bring ideas to life.

Yet, for many creative professionals, the new technology fell short. Adobe SVP of Design Eric Snowden remarked,

There were artists who found GenAI exciting, but who likened the experience of generating images with the technology to that of a slot machine: ‘I push a button and I have no idea what’s going to come out. I have no control. I’m not a poet or writer. These models expect I’m going to take this picture I have in my head and turn it into a text prompt that accurately describes what I want. That’s not how I work. If I need to make a tweak, it just generates all over again and it’s usually a different image.’

Others held distaste for the new technology altogether, saying the lack of humanity in the creation process led to outcomes lacking soul and meaning. They claimed GenAI art was inauthentic and little more than copy-and-pasted work of previously produced human art.

While technology watchers and early adopters were tracking, and often debating, developments in GenAI image generation, the broader creative community’s interest surged in the summer of 2022. A pivotal moment came in August that year, when a Midjourney-generated piece of art won first place at the Colorado State Fair’s fine arts competition.³⁵ The winner later disclosed his use of GenAI, sparking controversy among fellow artists. Critics saw it as a harbinger of automation encroaching on

^a A GPU, or, graphic processing unit, is made of billions of microscopic transistors that control the flow of electricity and are used to create images and videos. GenAI models require thousands of GPUs to train and the number and cost of these GPUs increases with model complexity.

an inherently human domain. One social media user lamented, “We’re watching the death of artistry unfold right before our eyes.”³⁶ (See **Exhibit 11** for winning entry.)

Chief among artists’ concerns was the threat of lost income due to replacement by other artists using GenAI technology or by GenAI images outright. They fretted that those without training, credentials, or years in the profession could enter the field and steal their jobs. They also worried that their art could be used for training the models without their consent, compensation, or control. “And worse, they felt they had no recourse,” noted Snowden. Not only was their work used as training, but the models could be used to mimic an artists’ unique style. This fear was voiced by concept artist R.J. Palmer, who was dismayed to see open-source tools like Stable Diffusion suggesting prompts based on his specific style. “When they’re feeding work from living, working artists who are struggling as it is, that’s just mean-spirited,” he remarked.³⁷ Meanwhile, illustrator Greg Rutkowski discovered that over 90,000 suggested prompts invoked his name—far more than those of many classical painters like Rembrandt or Picasso. Initially pleased by the prospect of wider exposure, Rutkowski realized it muddled his creative identity online, “It’s been just a month. What about in a year? I probably won’t be able to find my real work out there because [the Internet] will be flooded with AI art.”³⁸

The Debate: How to Approach GenAI for Content Creation

Adobe had been leveraging AI to solve creatives’ pain points for years. As early as 2017, Jon Brandt of Adobe Research voiced how AI could revolutionize the creative process. At a conference that year, he remarked, “We are here to invent creative AI and transform the world through machine learning... That said, figuring out how to make it work takes time.” In what would prove a prescient remark, Brandt imagined a world where “you could simply tell the tool what you want, and it becomes an intelligent assistant that handles the tedium, without disrupting your creativity.” He concluded that with AI, Photoshop would evolve “from just a tool into your creative assistant.”

With this bold vision in mind, the Adobe team spent the next several years building AI-powered tools, including one that enabled facial editing in Photoshop. This innovation leveraged pioneering work in GANs by Alexandru Costin, Head of the Applied Research Group. A pivotal moment in Adobe’s AI journey came in 2019, when Gavin Miller, Head of Research, organized an offsite for the research team to explore AI’s potential in creative applications. During this deep dive, the team penned a white paper envisioning the next generation of Adobe’s tools—where users could simply talk to the computer rather than using a task bar. “We started to dream up a vision to reinvent the creative process. We started to see the future,” Costin recalled.

The Inflection Point

Several executives pointed to OpenAI’s release of DALL-E 2 in spring 2022 as a watershed moment. Miller commented, “The first version of DALL-E felt like nothing more than an interesting curiosity, and we had no sense of how rapidly image generation technology would improve. But when DALL-E 2 came out, it was a step-change from the first version and it happened so quickly. That’s when I realized we needed to get on the train.” Wadhvani concurred, “I saw the first avocado chair iteration back when I was working at Greylock. Sam Altman used to work out of their office. Right away I knew, this was going to be big and that we would need to dive in. The next iteration made it that much more real.”

Having kept a finger on the pulse of developments in AI alongside his team, in summer 2022, Narayan kicked off a series of meetings to formally assess the situation and craft a plan of action. Participants included senior leaders from research, applied research, product, design, marketing, and

strategy. Wadhwani recalled, “The reason is that GenAI is a little bit like alchemy. There’s a deep science to it, but there’s a lot of experimentation and testing and a little salt over the shoulder that results in the thing looking right. We needed all perspectives in the room.”

A few small early rumblings inside the company suggested that Adobe should not get involved in the new space given the reaction from some in the creative community. Others suggested a wait-and-see approach. “One option was to have stuck our head in the sand or to have decided that the technology was never going to get there,” summarized one executive. Those in this camp felt that the images made by existing models—often rudimentary, grainy or with stylized personalities—were not ready for the exacting standards of Adobe’s professional creative customers who sought commercially-ready, high-quality imagery and pixel-level control. Still others, who saw the technology as promising, expressed concern that it might be “too late to catch up” to existing players.

Nonetheless, several respected voices inside the company felt it was necessary to dive into the space. Recalled Costin, “We needed some time to convince folks internally that this was the next big thing. And that we would need to take a leap of faith and dive in.” Malte Bernholz head of Corporate Strategy noted, “It quickly became clear we were facing a once-in-a-lifetime technology shift.”

Early Decision: Build, Buy, or Partner?

Questions emerged around how exactly Adobe would undertake this leap. Over a series of meetings, executives debated whether the company should partner with or even possibly acquire one of the existing GenAI image model creators. One executive recalled, “Advocates for partnering argued it could allow us to stick to our knitting, which had always been the tooling.” The temptation to partner with an existing player was strong as it would speed time to market. This strategy aligned with concerns voiced by skeptics who questioned whether Adobe had the right talent to undertake the heavy lift and high cost required in training a model. “We were hearing about how expensive it was to train LLMs like GPT. How much would training an image model be?” asked one executive.

However, several leaders pushed back on partnering. Said one,

Look, those other companies are chasing a different dream than we are. Some are chasing AGI (Artificial General Intelligence) as an end-goal, others are a bunch of academics coming together to build cool technology. Image generation is secondary to those players. Their goals are radically different from ours. They’re not tightly intertwined with the creative community like we are. We’ll not see eye-to-eye.

Another noted, “Some of those companies had a reputational stain given how they trained their models. Partnering might bring our customers to our doors with pitch forks. Heck, our own employees might join them.” One leader asserted, “If we do this, we need to control it ourselves and own our own destiny.”

Could Adobe build a model that matched that of players who were solely focused on this area and had the talent and funding to continuously innovate? Recalling those heated discussions, several leaders pointed to the perspective held by Gavin Miller. “He told me with incredible conviction—we can build our own. And not only that, but that ours would eventually match or exceed existing models,” recalled Narayan.

Those embracing the “build” vision were bolstered by Adobe’s human capital. Its deep bench of technical and research capabilities had close ties to academia, had published multiple white papers on AI, and had served in leadership roles in a key industry group on computer graphics and technical

interfaces. Adobe's in-house design team was another asset. "We were all Adobe customers before joining Adobe, so we understand our users deeply. In many ways, our team is Customer Zero. We're a good first litmus test to understand what customers want," noted Snowden. He added, "Our user research team was already talking to customers about their perceptions of the existing models. Our product equity team makes sure our products represent people from all walks of life equally. We could bring that lens to building a model that represented everyone."

Another tailwind in Adobe's favor was its internal compute power that had meaningfully ramped starting in mid-2021. Wadhwani recalled, "Gavin came to me pointing out that we were falling behind in GPUs per employee in the research area and that to be able to continue to hire the best talent, we would need to increase that spend sizably."

Adobe also had one of the world's largest curated libraries of high-quality, royalty-free stock images, videos, and illustrations. The platform originated from Adobe's acquisition of Fotolia and had grown to over 300 million assets contributed by a global community of creators. Adobe generated licensing revenue through asset purchases and subscriptions, while remitting royalties to contributors based on downloads of their images. Adobe Creative Cloud users were able to search for, license, and incorporate these assets into their projects from within the Adobe ecosystem. In addition, Adobe could draw upon decades of deep expertise and data across creative workflows in order to build its own model.

CFO Dan Durn began to contemplate how the move would impact financials, specifically profitability. Apart from training and ongoing variable inference costs, fortunately, there would be no physical infrastructure, as Adobe would rely on their cloud partnership with AWS for compute capacity. It was clear that investors were nervous as these new technologies emerged and some questioned what the future would look like, not only for Adobe as a company, but for its creative customers. He recognized that he had to articulate the vision in a way that brought investors along on the company's journey. "I need to help investors see the long-term value of this move, and not focus so much on the short term," he recalled.

As the team plowed into building its own model, Narayen maintained a careful eye on the industry. What milestones would the team use to benchmark and judge the quality of the model Adobe would build? How might he hedge his bets along the way without rocking the team's confidence?

Decisions around Training Adobe's Model

From the outset, Adobe leaders recognized that building a GenAI image model entailed a fundamental choice: to train exclusively on content that Adobe had permission to use or to cast a much wider net by mining publicly available data from the web. The stakes were high. Adobe had cultivated over four decades of trust with creative professionals and a misstep risked alienating them. "It was well-known that the available GenAI image models had been trained by scraping data from the Internet. Content owners claimed that training on their data was a violation of their rights because it is essentially unauthorized copying of their material, in violation of the Copyright Act," commented Andrew Savage, VP Legal Digital Media. However, he noted, "Setting aside any questions around implicit or explicit permission to copy data from the Internet, most of the companies who have trained in that way also claim that what they've done should be considered fair use.^b"

^b The U.S. Copyright Act typically evaluated fair use based on: the purpose and character of the use, the nature of the original work, the amount and substantiality of the portion used, and the effect of the use on the market for the original work.

While the legality of mining data from the Internet to train generative AI models for image creation was unclear, past legal cases offered some guidance. For instance, courts had ruled that using thumbnail images of art in search results was permissible because the thumbnails were transformative, served a new and beneficial purpose, and did not directly harm the market for the original works. These decisions suggested that fair use might apply when images are used in a fundamentally different way that provides societal value, though how these principles applied to AI training remained unsettled.³⁹ Adobe Senior Director of IP, Advertising, and AI Policy J. Scott Evans summarized, “It may be a decade before we know, and it might go all the way to the Supreme Court.” Until then, other companies would be moving fast. As Bernholz noted, “The history of innovation is often at the border of legality. Just look at Uber.”

A significant contingent within Adobe believed that a model trained only on content Adobe had permission to use should be the path. “We did not want Adobe to go about training its model the way everyone else was doing it,” asserted Samantha Warren of the Adobe Design Team, who had previously worked as a creative freelancer. Such a path could leverage Adobe’s massive Stock catalogue. A decade-plus-old provision in Stock contributor terms allowed Adobe to use Stock images to create new products, services, and features. Moreover, these Stock images had undergone extensive AI and human moderation to strip them of IP and inappropriate content. A model trained exclusively on commercially safe content could also leverage content in the public domain (material with expired copyright, that which was never copyrighted, and that which was part of the Creative Commons).^c Moreover, Adobe could pay to license additional materials. Noted Bernholz,

A model trained on commercially safe content would allow added security for our customers. We could offer them indemnification whereby if something IP-protected ended up in work produced in a model we made, we would protect them legally from being sued. Such a model would not be contaminated with brand assets from other companies.

One Adobe executive noted that some large enterprise clients and agencies would simply “never touch” GenAI unless they could trust it was commercially safe.

Proponents stressed that such a model—untainted by brand assets—aligned with Adobe’s values, especially around respecting creative rights while ensuring commercial viability to enterprise customers. Evans illustrated the distinction: “If you ask one of the existing models to give you a picture of Batman, they serve it up. But if you asked a commercially safe model to do so, it would return an image of the animal bat alongside a man, as it would be unaware of Batman from its training data.” Costin clarified that creating a model trained only on commercially safe content did not mean that companies could not bring their own IP into the mix. He was already musing about an approach to allow enterprises and agencies to leverage their brand’s assets while using an Adobe model.

For many at Adobe, a model trained on exclusively commercially safe content presented precisely the level of caution customers expected. Moreover, such a model might be helpful given that Adobe played on an international field; they needed to be sensitive to regulatory and legal environment in different geographies. “Ultimately, we wanted to be viewed as the adults in the room when it came to building a model,” said Costin.

^c Creative Commons is a nonprofit that provides a framework of free, standardized licenses for creators to use when sharing their work. These licenses allow creators to grant varying levels of permission for others to use, modify, distribute, or build upon their works.

Still, others argued for a model trained on all available content from the Internet. “If everyone else is making commercially unsafe models and scraping the web, then are we shooting ourselves in the foot if we don’t? If we assume more data leads to better model outcomes, would we ever be able to match the quality of the existing models if we go the commercially safe route?” quipped one executive. “Tapping publicly available data might yield more robust outputs and richer creative possibilities, which could attract customers less concerned about content provenance,” they continued. “We serve many segments, including casual creators,” observed Snowden. “For them, it’s all about the outcome. They crave simplicity, and they’re less anxious about the source of the training data.”

Even within the creative professionals segment, might there be those who wished to access copyright protected materials for certain stages of their work? For instance, as Chief Strategy Officer Scott Belsky noted, “If you look at how creatives ideate today, they are ripping pages from magazines and pulling images from Pinterest and Google to populate their idea boards. But, when it’s time to create, they shift to commercially safe tools.” Might Adobe wish to offer these creative professionals access to the existing models from within the Adobe ecosystem?

Belsky pointed to the company’s roots as an open ecosystem. “We’ve always been a platform-agnostic company.” In fact, Adobe was already leveraging third-party LLMs in other parts of the business for text-based processes, so he advocated for allowing third-party image models onto Adobe’s platform. He drew an analogy, “Models are like cameras. They generate content that you then edit before you use it. The output of the model is not your end product. So, even if customers use third-party image models, they will still use our products.” Belsky was sanguine about developments in the competitive landscape. “Whenever another image model gets better, I want us to be happy,” he commented. Just how to incorporate third-party image models was an open item on the leadership team’s meetings. One executive noted, “If we let users access other GenAI image models on our platform, do we risk them going directly to those other models in the future? How might we live up to commercial-safety if output from these other models shows up in our ecosystem?”

How to Monetize

However Adobe decided to train its model, how it introduced and priced the tool mattered. Should it launch a standalone product? Or should the tool be interwoven into the software as a “superpower” that allowed users to go beyond text prompts to highly integrated creative workflows? The answer depended on whether Adobe prioritized strengthening its installed user base or expanding to new audiences. Dan Egan from Corporate Strategy argued for the latter, in order to delight subscribers. “Our strength is in helping customer workflows,” he noted. “The real value will come from integrating this new technology into the products and services our customers know and love.”

Deciding on a monetization model posed unique challenges. Training a cutting-edge text-to-image model from scratch could require hundreds of millions of dollars in GPU hardware and cloud infrastructure—some estimating as high as \$100 million per training cycle. Licensing additional high-quality content could further increase expenses. Unlike traditional SaaS offerings, where marginal costs (after development) were usually low, maintaining a generative model involved ongoing inference costs, meaning the company incurred costs each time a user generated an image.

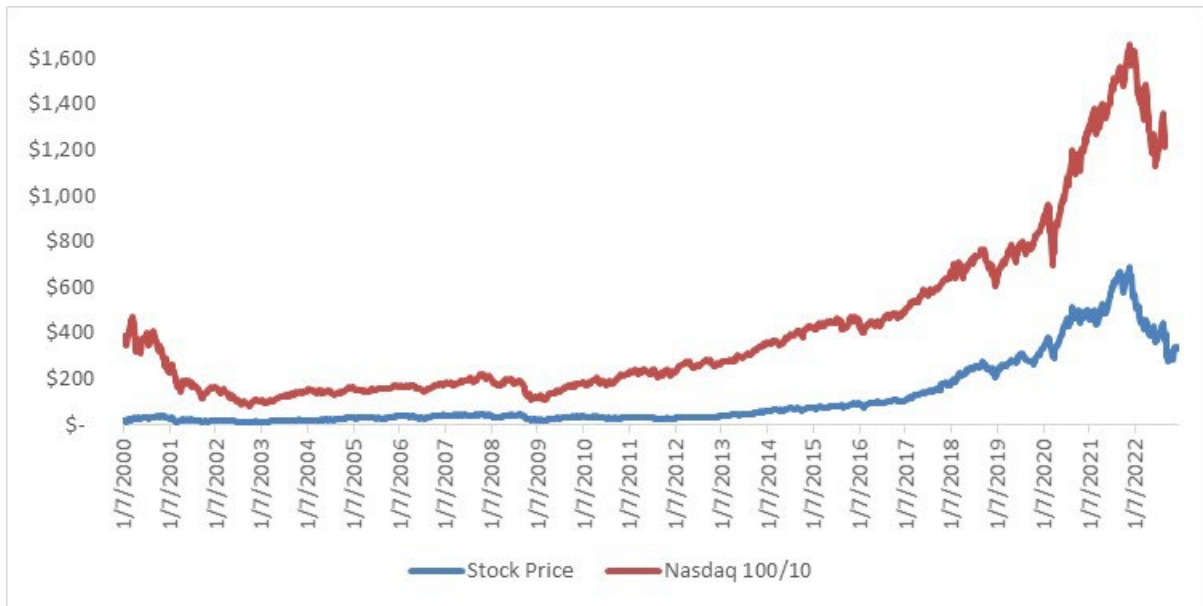
Adobe considered various pricing structures. Some GenAI players, like OpenAI, used a credit system where users could generate a certain number of images for free and then had to pay for additional images. Others, like Midjourney, offered tiered subscriptions. Stability AI had taken a different path, releasing Stable Diffusion in open-source and letting others bear the infrastructure costs. Adobe would need to consider whether it should meter usage, for example charging per image, or bundle AI features into existing subscriptions, risking higher expenses, but driving rapid adoption.

Enterprise customers might prefer site-wide licensing to integrate AI-driven workflows seamlessly, but this would require careful capacity planning and had some financial exposure.

A Sense of Urgency

As Narayen closed the news app on his phone, he reflected on Adobe's mission: to change the world through personalized digital experiences. Was GenAI a step on the way to changing the world of creativity? He found himself returning to a striking prediction by Scott Branson: "GenAI will simultaneously lower the floor and raise the ceiling for creativity." On one hand, it promised to democratize creativity, enabling newcomers with limited training to produce polished imagery; on the other, it would give professionals freedom to experiment and refine very rapidly, freeing them for higher-order creative tasks. This vision energized Narayen.

He acknowledged Adobe's enduring belief that technology cannot supplant creativity as an inherently human capability, and he remained convinced that the most effective solutions would enhance human creativity. However, Narayen could not help but reflect on the tension between this view of creativity and the brutal competitive realities of the technology industry. Could GenAI strengthen Adobe's position or would it usher in new competitors that might threaten its leadership? One thing seemed certain: speed mattered in this era of unprecedented innovation. Adobe's history had shown that moving decisively at key inflection points was vital. Yet no earlier transformation had cut so close to the essence of creativity. With the leadership meeting about to begin, Narayen opened his notebook, determined to chart a path that would embrace GenAI's extraordinary power while preserving Adobe's ethos as a champion of human-led design.

Exhibit 1 Adobe Stock Chart versus NASDAQ 100 (2000 to 2022)

Source: Created by casewriters using data from Google Finance, <https://www.google.com/finance/quote/ADBE:NASDAQ?hl=en>, accessed 1/21/25.

Note: NASDAQ 100 figure has been scaled down by a factor of 10 for ease of comparison with the Adobe price per share.

Exhibit 2a Revenue Breakdown by Segment 2018 to 2022 (in billions USD)

	2018	2019	2020	2021	2022
Digital Media	6.36	7.71	9.23	11.52	12.84
Digital Experience	2.54	3.21	3.40	3.87	4.42

Source: Company documents.

Exhibit 2b Consolidated Balance Sheet Year End 2013 to 2022 (in millions USD)

	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
ASSETS										
Current Assets:										
Cash and Equivalents	835	1,117	877	1,101	2,306	1,643	2,650	4,478	3,844	4,236
Short-term investments	2,339	2,622	3,111	3,750	3,514	1,586	1,527	1,514	1,954	1,860
Total Cash & ST Investments	3,174	3,739	3,988	4,851	5,820	3,229	4,177	5,992	5,798	6,096
Trade receivables, net of allowances for doubtful accounts	600	592	672	833	1,218	1,316	1,535	1,398	1,878	2,065
Prepaid expenses and other current assets	272	271	161	156	210	312	783	756	993	835
Total current assets	4,046	4,602	4,821	5,840	7,248	4,857	6,495	8,146	8,669	8,996
Property and equipment, net	660	785	787	816	937	1,075	1,293	1,517	1,673	1,908
Operating lease right-of-use assets, net	-	-	-	-	-	-	-	487	443	407
Goodwill	4,772	4,721	5,367	5,406	5,822	10,581	10,691	10,742	12,668	12,787
Other intangibles, net	605	470	510	414	386	2,069	1,721	1,359	1,820	1,449
Deferred income taxes	-	-	-	-	-	-	-	1,370	1,085	777
Other assets	297	208	241	140	144	187	562	663	883	841
Total assets	10,380	10,786	11,726	12,697	14,536	18,769	20,762	24,284	27,241	27,165
LIABILITIES										
Current Liabilities:										
Trade payables	62	68	93	88	114	186	209	306	312	379
Accrued expenses	657	684	678	740	994	1,163	1,399	1,422	1,736	1,790
Debt	15	603	-	-	-	-	3,149	-	-	500
Deferred revenue	776	1,098	1,434	1,946	2,406	2,916	3,378	3,629	4,733	5,297
Income taxes payable	10	24	6	38	14	36	56	63	54	75
Operating lease liabilities	-	-	-	-	-	-	-	92	97	87
Total current liabilities	1,526	2,494	2,213	2,811	3,527	4,301	8,191	5,512	6,932	8,128
Long-term Liabilities:										
Debt	1,499	911	1,907	1,892	1,881	4,125	989	4,117	4,123	3,629
Deferred revenue	53	57	51	69	89	138	123	130	145	117
Income taxes payable	133	126	256	184	173	644	616	529	534	530
Operating lease liabilities	-	-	-	-	-	-	-	499	453	417
Other liabilities	445	421	297	315	405	199	313	233	257	293
Total liabilities	3,656	4,009	4,724	5,272	6,075	9,407	10,232	11,020	12,444	13,114
SHAREHOLDER'S EQUITY										
Common stock	61	61	61	61	61	61	61	-	-	-
Additional paid-in-capital	3,393	3,778	4,184	4,616	5,082	5,685	6,504	7,357	8,428	9,868
Retained earnings	6,929	6,924	7,253	8,115	9,574	11,816	14,829	19,611	23,905	28,319
Accumulated other comprehensive stock at cost	46	(8)	(169)	(173)	(112)	(148)	(188)	(158)	(137)	(293)
Treasury Stock	3,643	(3,919)	(4,268)	(5,132)	(6,084)	(7,990)	(10,615)	(13,546)	(17,399)	(23,843)
Total stockholder's equity	6,725	6,776	7,001	7,424	8,460	9,363	10,530	13,264	14,797	14,051
Total liabilities and stockholder's equity	10,380	10,786	11,726	12,697	14,535	18,770	20,762	24,284	27,241	27,165

Source: Company documents.

Exhibit 2c Consolidated Income Statement Year End 2013 to 2022 (in millions USD)

	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Revenue										
Subscription	1,138	2,077	3,224	4,585	6,134	7,604	9,634	11,626	14,573	16,388
Product	2,470	1,628	1,125	800	707	622	648	507	555	532
Services and other	447	443	446	469	461	804	889	735	657	686
Total revenue	4,055	4,147	4,796	5,854	7,302	9,030	11,171	12,868	15,785	17,606
Cost of Revenue										
Subscription	278	335	409	462	623	574	926	1,108	1,374	1,646
Product	138	97	90	69	57	46	40	36	41	35
Services and other	170	190	245	289	330	575	707	578	450	484
Total cost of revenue	587	622	744	820	1,010	1,195	1,673	1,722	1,865	2,165
Gross Profit	3,469	3,525	4,051	5,035	6,292	7,835	9,498	11,146	13,920	15,441
Operating Expenses										
Research and development	827	844	863	976	1,224	1,538	1,930	2,188	2,540	2,987
Sales and marketing	1,620	1,652	1,683	1,910	2,198	2,621	3,244	3,591	4,321	4,968
General and administrative	520	543	532	576	625	745	881	968	1,085	1,219
Amortization of intangibles	52	52	69	79	77	91	175	162	172	169
Total operating expenses	3,046	3,112	3,148	3,540	4,124	4,995	6,230	6,909	8,118	9,343
Operating Income	423	413	903	1,494	2,168	2,840	3,268	4,237	5,802	6,098
Non-operating Income (expense):										
Interest expense	(68)	(60)	(64)	(70)	(74)	(89)	(157)	(116)	(113)	(112)
Investment gains (losses), net	(4)	1	1	(2)	8	3	52	13	16	(19)
Other income (expense), net	5	8	34	14	36	40	42	42	-	41
Total non-operating expenses, net	(67)	(51)	(29)	(58)	(30)	(46)	(63)	(61)	(97)	(90)
Income before income taxes	356	361	874	1,435	2,138	2,794	3,205	4,176	5,705	6,008
Provision for (benefit from) income taxes	66	93	244	266	444	203	254	(1,084)	883	1,252
Net income	290	268	630	1,169	1,694	2,591	2,951	5,260	4,822	4,756
Basic net income per share	\$ 0.58	\$ 0.54	\$ 1.26	\$ 2.35	\$ 3.43	\$ 5.28	\$ 6.07	\$ 10.94	\$ 10.10	\$ 10.13
Shares used to compute basic net income per share	501	498	499	498	494	491	486	481	477	470
Diluted net income per share	\$ 0.56	\$ 0.53	\$ 1.24	\$ 2.32	\$ 3.38	\$ 5.20	\$ 6.00	\$ 10.83	\$ 10.02	\$ 10.10
Shares used to compute diluted net income per share	513	508	507	504	501	498	492	485	481	471

Source: Company documents.

Exhibit 3a Creative Cloud Products and Services

Product	Features
<i>Adobe Photoshop and Lightroom</i>	• <i>Photoshop</i>: A widely used digital imaging and design application, available on desktop, iPad, and via a web-based beta. Provides advanced editing, effects tools, and neural filters. • <i>Lightroom</i>: A cloud-based photo service for editing, organizing, storing, and sharing images across multiple devices, offering features such as Select People, Select Objects, and Content-Aware Remove. • A Photography Plan bundles both Photoshop and Lightroom for photographers and enthusiasts.
<i>Adobe Illustrator and Fresco</i>	• <i>Illustrator</i>: An industry-standard vector graphics application for creating illustrations and digital graphics on desktop and iPad. Used for print, web, interactive, video, and mobile media. • <i>Fresco</i>: An illustration app (free or premium) on iPhone, iPad, Windows devices, and Wacom MobileStudio devices. Combines vector, raster, and Live Brushes (powered by Adobe Sensei) for a natural painting and drawing experience.
<i>Adobe InDesign</i>	• A design and layout application for print and digital media. Often used to create books, eBooks, magazines, posters, and interactive PDFs. Integrates with Photoshop, Illustrator, Acrobat, and Adobe Stock, and enables creating and publishing apps for iOS, Android, and Amazon devices.
<i>Adobe Premiere Pro and Premiere Rush</i>	• <i>Premiere Pro</i>: A professional nonlinear video editing application for filmmakers, TV editors, YouTubers, and videographers. Supports various media formats and integrates with Frame.io for review and approval. • <i>Premiere Rush</i>: An all-in-one video editing app (free or premium) with simplified editing tools for quick content creation and direct sharing to platforms like YouTube, Instagram, and TikTok.
<i>Adobe Express</i>	• A web-based platform and mobile app focused on easy content creation for first-time creators and communicators. Offers guided tools, one-click solutions, templates, and a Content Scheduler for planning and publishing social media posts. Includes free and premium features.
<i>Adobe Stock</i>	• A collection of millions of high-quality, royalty-free photos, vectors, illustrations, videos, templates, audio, and 3D assets. Integrated with many Creative Cloud apps for streamlined asset discovery and licensing, with free and paid options.
<i>Substance 3D</i>	• An ecosystem of desktop apps—including Stager, Painter, Sampler, Designer, and Modeler—for 3D design and texturing. Allows users to assemble 3D scenes, texture assets with brushes and Smart Materials, and import professional 3D textures from the Substance 3D Assets library. Modeler enables VR and desktop-based sculpting.
<i>Adobe After Effects</i>	• A motion graphics and visual effects application for cinematic titles, effects, and animations. Integrates with Premiere Pro, Photoshop, Illustrator, and other tools. Includes Frame.io integration for real-time collaboration, scene edit detection, and 3D Extended Viewer.
<i>Frame.io</i>	• A cloud-native video collaboration platform with real-time uploads, review, and approval. Integrates directly with Premiere Pro and After Effects for frame-specific commenting. Camera to Cloud automatically uploads footage from on-set devices for immediate editing and feedback.
<i>Behance</i>	• A social community where creators can showcase work and discover others' projects online. Integrates with Adobe Portfolio for building customizable websites, and supports live-streaming from Creative Cloud applications.

Source: Company documents.

Exhibit 3b Creative Cloud Customer Breakdown

Segment	Members
Business Professionals	Social Media Teams, Sales Reps, HR Teams, Communication Teams, Small Business Owners
Consumers	Students, PTAs, Personal use cases like travel, invitations, announcements
Creative Professionals	Graphic Designers, Photographers, Videographers, Illustrators, 3D Artists
Creators	Social Media Influencers, Solopreneurs, Travel Bloggers, Gig Workers
Marketing Professionals	Brand Managers, Channel Marketers, Campaign Strategists, Performance Marketers, Social Marketers, Data Analysts, Marketing Ops

Source: Company documents.

Exhibit 4a Document Cloud Products and Services

Product	Features
<i>Adobe Acrobat</i>	- A leading application for creating, converting, and editing PDFs. Provides secure, reliable, and compact PDF document creation, plus collaborative workflows with commenting, editing, sharing tools, and direct integration with Acrobat Sign. - Also available as part of the Creative Cloud All Apps subscription.
<i>Acrobat Web</i>	A web-based suite of PDF tools that allows users to edit, comment, convert, organize, and sign PDF documents directly in the browser. Offers quick, one-click features, including both free and premium capabilities.
<i>Adobe Acrobat Reader</i>	- A free software tool for viewing, annotating, and printing PDF documents on desktop and mobile. Includes options to create, edit, export, combine, share, and collaborate on PDFs, along with "Liquid Mode" for easy reading on mobile devices. - Integrates with Adobe Document Cloud as well as third-party storage services.
<i>Adobe Scan</i>	A free mobile app that captures paper documents as images and transforms them into full-featured PDFs using Document Cloud services. Enables quick scanning, sharing, and storage on any mobile device.
<i>Adobe Acrobat Sign</i>	- A cloud-based e-signature service that allows secure electronic sending, signing, and tracking of documents on any device. - Integrates with enterprise systems through APIs, Adobe Experience Manager Forms, and Advanced Workflows for Acrobat Sign, enabling seamless web and mobile e-signature experiences.

Source: Company documents.

Exhibit 4b Document Cloud Customer Segment

Segment	Members
Business Professionals	Social Media Teams, Sales Reps, HR Teams, Communication Teams, Small Business Owners
Consumers	Students, PTAs, Personal use cases like travel, invitations, announcements

Source: Company documents.

Exhibit 5a Digital Experience Products and Services

Product	Features
<i>Adobe Experience Manager</i>	A content management solution for websites, mobile apps, and other digital touchpoints, used for content authoring, asset management, digital forms, and personalized content delivery
<i>Adobe Analytics</i>	A data analytics platform for tracking user interactions across multiple channels, providing real-time segmentation, attribution modeling, and predictive insights through interactive dashboards
<i>Adobe Target</i>	A personalization and testing engine that tailors user experiences, enabling A/B testing, multivariate experiments, and AI-driven recommendations
<i>Adobe Audience Manager</i>	A data management platform that aggregates audience data from various sources, creating high-value segments for ad targeting and personalization across channels
<i>Adobe Campaign</i>	A cross-channel campaign management tool for orchestrating messaging, supporting email, SMS, and push notifications, with data-driven automation and personalized workflows
<i>Marketo Engage</i>	A marketing automation solution geared toward lead management and account-based marketing, facilitating personalized campaigns, scoring models, and sales-marketing alignment
<i>Adobe Commerce</i>	An e-commerce platform enabling omnichannel storefront creation, managing product catalogs, payment integrations, and shopping experiences at scale
<i>Adobe Workfront</i>	A work management platform designed for project planning and collaboration, providing workflow automation, asset review, and progress tracking
<i>Adobe Journey Optimizer & Real-Time CDP</i>	Solutions for unified customer profiles and real-time journey orchestration, leveraging AI to personalize engagement across channels using data from multiple sources

Source: Company documents.

Exhibit 5b Digital Experience Customer Breakdown

Segment	Members
Creative Professionals	Graphic Designers, Photographers, Videographers, Illustrators, 3D Artists
Marketing Professionals	Brand Managers, Channel Marketers, Campaign Strategists, Performance Marketers, Social Marketers, Data Analysts, Marketing Ops

Source: Company documents.

Exhibit 6 Sample Content Supply Chain Workflow

Stage	Description	Key Adobe Tools
Workflow & Planning	Teams align on campaign goals, resources, and customer data	Workfront
Creation & Production	Creative teams produce assets and content at scale	Creative Cloud, Express, Frame.io
Asset Management	Stakeholders collaborate, edit, and ensure compliance	Experience Manager
Delivery & Activation	Personalized campaigns and assets launched across channels	Native connectivity to any Experience Cloud delivery application; integrates with 3rd party or custom delivery apps
Reporting & Insights	Performance tracked, insights generated, and content refined	Content analytics

Source: Company documents.

Exhibit 7a Select Competition in Digital Media

Competitor	Company Value	Estimated Annual Revenue	Key Products/Offerings	Competitive Advantages
Autodesk	\$40.32 billion market cap	\$4.4 billion	AutoCAD, Maya, Inventor	• 3D design, engineering, and entertainment software • Industry-standard tools for architecture, engineering, and manufacturing
Corel	\$1 billion acquisition value by KKR	\$250 million	CorelDRAW, Corel Painter, Corel Photo-Paint	• Graphics, photo editing, and digital painting software • Intuitive tools and features for creative professionals
Serif (Affinity)	Private, undisclosed	\$60 million	Affinity Designer, Photo, Publisher	• Professional-grade design software for graphic designers, photographers, and publishers • One-time purchase option

Source: Compiled by casewriters.⁴⁰**Exhibit 7b** Select Competition in Digital Experience

Competitor	Market Cap	Annual Revenue	Key Products/Offerings	Competitive Advantages
Google	\$1.2 trillion	\$282 billion	Google Analytics, Google Ads, Google Marketing Platform	Wide range of marketing and analytics tools, including web analytics, advertising, and marketing automation
HubSpot	\$15 billion	1.73 billion	HubSpot Marketing Hub	Marketing automation platform that helps businesses attract visitors, convert leads, and close customers
Microsoft	\$1.9 trillion	\$198 billion	Microsoft Dynamics 365	Suite of business applications, including CRM and marketing capabilities
Oracle	\$220 billion	\$42.4 billion	Oracle Customer Experience (CX) Cloud Suite	Comprehensive suite of cloud applications, including CRM, marketing automation, and data analytics
Salesforce	\$135 billion	\$26.5 billion	Salesforce Sales and Marketing Cloud	Market leader in CRM space and has AI-enabled sales and marketing automation

Source: Compiled by casewriters.⁴¹

Exhibit 8 Select Images using Harold Cohen's "Aaron"

Drawing, computer-generated, with hand colouring, by Harold Cohen, 1974.
© Victoria and Albert Museum, London



Drawing, computer-generated, with hand colouring, by Harold Cohen, 1977-1982.
© Victoria and Albert Museum, London

Excerpt from Cohen's 1982 work, *How to Make a Drawing*:

"Aaron was clearly not a tool in an orthodox sense. It was closer to being a sort of assistant, if the need for an human analogue persists, but not an assistant which could learn what I wanted done by looking at what I did myself, the way any of Rubens' assistants could see perfectly well for themselves what a Rubens painting was supposed to look like. This was not an assistant which could perform any better for having done a thousand drawings, not an assistant which could bring anything approximating to a human cognitive system to bear on the production of drawings intended for human use.

We are living on the crest of a cultural shock-wave of unprecedented proportions, which thrusts a new kind of entity into our world: something less than human, perhaps, but potentially capable of many of the higher intellectual functions — it is too early still to guess HOW many — we have supposed to be uniquely human. We are in the process of coming to terms with the fact that "intelligence" no longer means, uniquely, 'human intelligence.'"

Source: Images available via Kate Vass, "Harold Cohen: 'Once upon a time there was an entity named Aaron,'" *Kate Vass Galerie*, April 30, 2024, <https://www.katevassgalerie.com/blog/harold-cohen-aaron-computer-art>, accessed January 18, 2025. Excerpt from Harold Cohen, "How to Make a Drawing," December 17, 1982, available via <https://www.aaronshome.com/aaron/publications/how2make.pdf>, accessed May 2025.

Exhibit 9 Advancements DALL-E to DALL-E 2

Source: Images available via Sarah Guo blogpost, “Thinking (creatively) with the help of machines,” April 7, 2022, <https://sarahguo.com/blog/dalle2>, accessed January 20, 2025.

Exhibit 10 Sample Midjourney Image, August 2022

Source: Kyle Kukshtel blogpost, "Midjourney Changed my Mind on AI-Generated Art," June 24, 2022, <https://kylekukshtel.com/midjourney>, accessed January 20, 2025.

Note: Image generated by Midjourney using the prompt, "A Game Developer Writing a Blog Post."

Exhibit 11 Colorado Art Fair Winner, Theatre de Opera Spatial

Source: Jason Allen, "Théâtre D'opéra Spatial," 2022, available via https://commons.wikimedia.org/wiki/File:Th%C3%A9%C3%A2tre_D%E2%80%99op%C3%A9ra_Spatial.jpg, accessed April 2025.

Note: According to video game artist and contest entrant Jason Allen, he spent 80 hours working on this piece of art. He entered the text prompts into Midjourney, which generated 800 images, from which he selected his favorite three. He reportedly used Photoshop to edit the images before entering them to the art competition.

Endnotes

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